

Joint inversion of gravity and magnetic data using correspondence maps with application to geothermal fields

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SUMMARY

Current geophysical joint inversion methods can be roughly divided into structurally or property driven. While structural coupling plays a key role in fully heterogeneous environments, it is blind to the actual physical properties, which limit its application. On the other hand, approaches based on direct property-correlations promise strong coupled models using petrophysical linkages derived from rock physics or through the search for cross-property relationships in the objective function directly, for example, fuzzy-clusters or correspondence maps. We apply a correspondence maps (CM) methodology to the joint estimation of density and magnetization in 3-D environments using gravity and magnetic data. CM joint inversion has yet to be applied to the specific combination of gravity and magnetic data, nor compared to other approaches like cross-gradient coupling or fuzzy-clustering. The CM searches for cross-property functional relationships; it has the potential to allow a gradual property variation that is not restricted to a predetermined number of homogeneous zones. We tested the algorithm using two illustrative synthetic examples to analyse the compositional and geometrical improvements and thoroughly investigate its capacity to recover a study model based on potential features in a geothermal scenario using polynomials. We investigated the case of using lower and higher polynomials than required for recovering the second synthetic model. The nature of the inversion suggested instability when high-grade polynomials were used, which was successfully overcome by gradually increasing the number of coefficients. The strategy was satisfactorily implemented to the field data.

We applied the algorithm over the Los Humeros geothermal field in Mexico and fit a quadratic relationship between density and magnetization that exhibits two main zones consistent with previous works. The first one contains relatively low density and high magnetization. The second zone contains low density and low magnetization values. In this last zone, we emphasize that an observed link between low density, low magnetization, and major fault zones with high relevance to geothermal exploration is resulting from joint inversion, only. We achieved an enhanced petrophysical and structural integrated subsurface imaging in synthetic and field cases at the expense of slightly increased RMS misfit value.

Key words: Gravity anomalies and Earth structure; Magnetic anomalies: modelling and interpretation; Joint inversion.

INTRODUCTION

Gravity and magnetic surveys are widely used in various geological settings. The possibility for low-cost regional surveys and their concurrent acquisition led to their joint application for mapping and discrimination of groundwater, thermal water, mineral or hydrocarbon targets (e.g. Baillieux *et al.* 2013; Martinez & Li 2015; Tontini *et al.* 2016). Despite this wide and long-standing use, there are only

a few quantitative strategies for the combined interpretation of gravity and magnetic data for 3-D targets based on geophysical inverse theory.

First combined inversions of both data types are achieved by reparametrizing the problem as finding depths to homogeneous (or quasi-homogeneous) compartments commonly divided in layers (e.g. Zeyen & Pous 1993; Fedi & Rapolla 1999; Gallardo-Delgado *et al.* 2003; Gallardo *et al.* 2005; Pilkington 2006; Fullagar *et al.*

2008; Jiang *et al.* 2008). These methods are highly popular due to their versatility to assimilate direct geological evidence in the form of contacts and faults within the models and their applicability to typical sedimentary basins (e.g. Mahmoodi *et al.* 2017; Christiansen *et al.* 2018). Fully heterogeneous models pose a more difficult problem since they require the assimilation of a meaningful linking function between the density and magnetization of the rocks. Depending on the chosen linking assumption, current joint inversion methods can be roughly divided into structurally or property driven (see discussions in Bedrosian 2007; Gallardo & Meju 2011; Moorkamp 2017).

In the first category, we may find examples of different structural constraints among many (e.g. Haber & Oldenburg 1997; Gallardo & Meju 2003; Musil *et al.* 2003; Zhdanov *et al.* 2012a; Molodtsov *et al.* 2013). In the case of potential fields such as gravity gradient and magnetics, one of the most used structural constraints is that based on the cross-gradient approach (e.g. Fregoso & Gallardo 2009; Moorkamp *et al.* 2011; Zhou *et al.* 2015). Structural coupling methods are widely applicable because they do not make strong assumptions about the geophysical relationships, but the result can be a moderate improvement in resolution compared to the individual inversions (Lelièvre *et al.* 2012; Moorkamp 2017; Gross 2019), and it is blind to the actual physical properties which limit its application. These drawbacks can be somehow alleviated by adopting property-driven approaches.

The earliest approaches based on direct property relationships incorporate petrophysical linkages derived from rock physics. Here, we can find many examples, especially among seismic and gravity data (e.g. Lees & Van De Car 1991; Nielsen & Jacobsen 2000; Afnimar *et al.* 2002; Onizawa *et al.* 2002; Jegen *et al.* 2009; Maceira & Ammon 2009; Coutant *et al.* 2012; Ars *et al.* 2019). In practice, these petrophysical linkages are either site-specific, highly uncertain or in existent. Therefore, some efforts for considering uncertainties when no *a priori* analytical relationships exist have been explored. For example, geostatistical links such as those of Bosch (1999) or fuzzy-cluster approaches (Paasche *et al.* 2006; Carter-McAuslan *et al.* 2014; Sun & Li 2016) have been applied successfully for gravity and magnetic data to 2-D (*cf.* Bosch & McGaughey 2001) and 3-D (Sun & Li 2017) targets. Fuzzy clusters approaches can be considered more appropriate than simplified petrophysical assumptions for some geological conditions (Heincke *et al.* 2017). However, they are currently limited in the number of homogeneous units (clusters) within the model, which is unknown or difficult to determine in geologically complicated areas. This has been somehow alleviated by selecting a particular model after testing different numbers of clusters (e.g. Sun & Li 2016). In a different approach, we may search for cross-property functional relationships in the objective function directly. This should allow a gradual property variation that is not restricted to a predetermined number of homogeneous zones (e.g. Lelièvre *et al.* 2012; Heincke *et al.* 2017; Carrillo & Gallardo 2018) and favours the detection of a larger number of thinner layers embedded in practical hydrocarbon or groundwater targets. The only reported application that incorporates a relationship for fully heterogeneous 3-D gravity and magnetic targets can be found in Kamm *et al.* (2015). However, they resource to petrophysical samples to constrain their enforced relationship. Carrillo & Gallardo (2018) have previously incorporated the correspondence maps (CM) methodology to magnetotelluric (MT) and gravity data in a 2-D scenario. This approach is useful when no *a priori* information about a direct link between properties is available, and we aim at seeking simultaneously two models and their functional relationship. In this work, we extend the use of the CM methodology

to estimate the 3-D distribution of density and magnetization by jointly inverting gravity and magnetic data. We fully investigate the approach by using two illustrative synthetic experiments: a linear and a parabolic relationship. We compare the results from separate and joint inversion and assess the CM's advantages and limitations. Finally, we apply the algorithm in the field data from a geothermal target in Los Humeros, México, and demonstrate that the CM approach is viable in real Earth applications.

INVERSION METHODOLOGY

To develop our 3-D gravity and magnetic joint inversion algorithm, we first implement a conventional forward formulation to calculate the gravity and magnetic fields for an ensemble of rectangular prisms. We then set an objective function for our 3-D joint inversion problem that incorporates the correspondence maps and adopts an iterative scheme for its computational implementation.

Forward modelling

Our basic 3-D model is formed by homogeneous rectangular prisms characterized by their density and magnetization (Fig. 1). We use the formula derived by Plouff (1977) to compute the vertical component of the gravity field caused by a rectangular prism with constant density. Similarly, we use the formulation derived by Bhattacharyya (1966) to compute the corresponding total magnetic field (T) of an individual rectangular prism. By superposition, the gravity and magnetic responses of the whole model (Fig. 1) are the added contribution of all the composing prisms.

Objective function

We define our objective function as finding the density model ($\mathbf{m}_1$), magnetization model ($\mathbf{m}_2$) and the coefficients ($\mathbf{a}$) of a polynomial function p that:

- i) fit the observed vertical gravity ($\mathbf{d}_1$) and magnetic ($\mathbf{d}_2$) data at a reasonable level,
- ii) are smooth,
- iii) resemble any preconceived geological model ($\mathbf{m}_{pr01}$, $\mathbf{m}_{pr02}$) and
- iv) correlate both parameters through a correspondence map function $p(\mathbf{m}_1, \mathbf{m}_2, \mathbf{a})$.

All these requirements are aggregated into the single objective function:

$$\begin{aligned} \phi = & (\mathbf{d}_1 - F_1(\mathbf{m}_1))^T \mathbf{C}_{dd1}^{-1} (\mathbf{d}_1 - F_1(\mathbf{m}_1)) \\ & + (\mathbf{d}_2 - F_2(\mathbf{m}_2))^T \mathbf{C}_{dd2}^{-1} (\mathbf{d}_2 - F_2(\mathbf{m}_2)) + \alpha_1 \|\mathbf{H}\mathbf{m}_1\|^2 \\ & + \alpha_2 \|\mathbf{H}\mathbf{m}_2\|^2 + (\mathbf{m}_{pr01} - \mathbf{m}_1)^T \mathbf{C}_{mpr01}^{-1} (\mathbf{m}_{pr01} - \mathbf{m}_1) \\ & + (\mathbf{m}_{pr02} - \mathbf{m}_2)^T \mathbf{C}_{mpr02}^{-1} (\mathbf{m}_{pr02} - \mathbf{m}_2) \\ & + (p_0 - p(\mathbf{m}_1, \mathbf{m}_2, \mathbf{a}))^T \mathbf{C}_{pp}^{-1} (p_0 - p(\mathbf{m}_1, \mathbf{m}_2, \mathbf{a})). \end{aligned} \quad (1)$$

They are weighted by statistical (covariance matrices) or empirical (α) measures. Here, F_1 and F_2 are the linear 3-D modelling operators which relate $\mathbf{d}_1$ to $\mathbf{m}_1$, and $\mathbf{d}_2$ to $\mathbf{m}_2$, respectively. $\mathbf{C}_{dd1}$ and $\mathbf{C}_{dd2}$ are the covariance matrices that consider the uncertainties in the corresponding observed data. $\mathbf{H}$ is a smoothness operator (in our case a first-order derivative) acting on $\mathbf{m}_1$ and $\mathbf{m}_2$, which is weighted by α_1 and α_2 , respectively. This term is largely used (i) to solve the ill-posed inverse problem in geophysics, especially severe

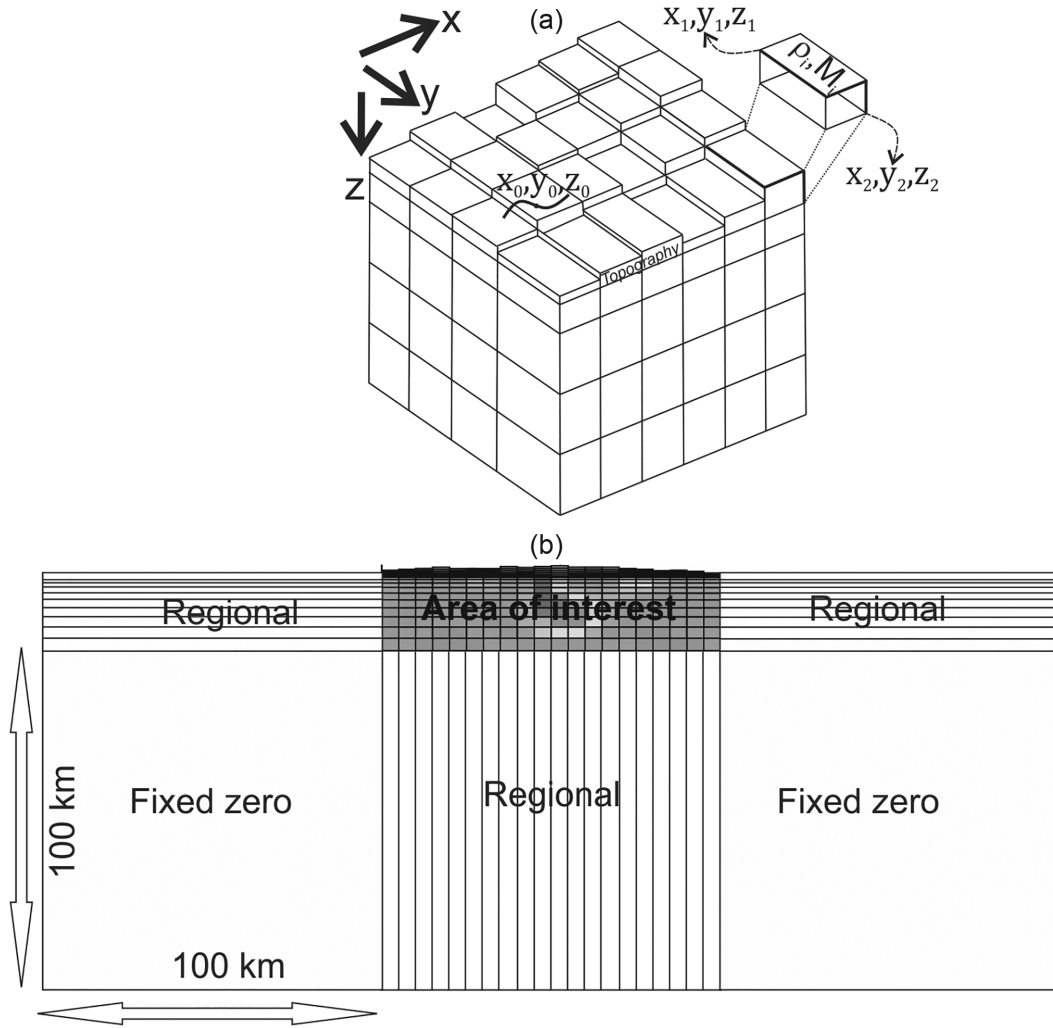


Figure 1. The discretized 3-D model of the subsurface is composed of a set of non-equally spaced rectangular prisms of constant density and magnetization (inset). The first layers of prisms are used to accommodate the topographic relief.

in the case of gravity and magnetic data in a full heterogeneous medium and (ii) to produce smoothly varying models, which are in many cases suitable to identify some regional geological structures and to avoid overinterpreting the data (e.g. Jupp & Vozoff 1975; Constable *et al.* 1987). $\mathbf{m}_{pr01}$ and $\mathbf{m}_{pr02}$ are any *a priori* density and magnetization models that can be used to guide our model search. For example, conceptual models elaborated from borehole, geological and geophysical data. $\mathbf{C}_{prm01}$ and $\mathbf{C}_{prm02}$ are the corresponding *a priori* models' covariances. The last term is the coupling term that relates density to magnetization. We aim at finding the coefficients ($\mathbf{a}$) of a polynomial function $\mathbf{p}$ that fits the estimated cross-property values ($\mathbf{m}_1, \mathbf{m}_2$) corresponding to the rectangular prisms. Therefore, $\mathbf{p}$ is an implicit function of $\mathbf{m}_1, \mathbf{m}_2$ and $\mathbf{a}$ given by

$$\mathbf{p}(\mathbf{m}_1, \mathbf{m}_2, \mathbf{a}) = \sum_{i=0}^q \sum_{j=0}^s a_{ij} m_1^i m_2^j, \quad (2)$$

where $\mathbf{m}_1$ and $\mathbf{m}_2$ are the density and magnetization parameters, $\mathbf{a}_{ij}$ contains the coefficients describing the polynomial for the 3-D distribution of density and magnetization for each prism in the space and q and s are the maximum power allowed for each parameter. $\mathbf{p}_0$ is the normalized targeted level set of the density–magnetization correspondence function, $\mathbf{p}_{calc}(\mathbf{m}_1, \mathbf{m}_2, \mathbf{a})$ is the correspondence map

function calculated for the actual distribution of density, magnetization and polynomial coefficients ($\mathbf{m}_1, \mathbf{m}_2, \mathbf{a}$) and, in the general case, it is a non-linear operator since we can use a polynomial of any order. $\mathbf{C}_{pp}$ is the preset covariance matrix of $\mathbf{p}$, mainly used to prescribe an acceptable level of density–magnetization correspondence. The readers are referred to Carrillo & Gallardo (2018) for more details.

Computational implementation

The optimal models and polynomial parameters are found through an iterative Gauss–Newton scheme. In this search procedure, all geological information (e.g. borehole data or previous geological models) are integrated into an *a priori* model by linking them to *a priori* density and magnetization values. The iterative process starts from two initial models $\mathbf{m}_{01}$ and $\mathbf{m}_{02}$. The gravity and magnetic responses are linear, and their sensitivity matrices remain constant during the process. In the general case, the polynomial equation is non-linear and their sensitivity matrices ($\frac{\partial \mathbf{p}}{\partial \mathbf{m}_{density}}, \frac{\partial \mathbf{p}}{\partial \mathbf{m}_{magnetization}}$ and $\frac{\partial \mathbf{p}}{\partial \mathbf{a}}$) are updated at each iterative step. In the following steps, we first update the models by solving the linearized system of equations and then calculate the new data response. The iteration is terminated

either at an acceptable RMS of data misfit, when the models are changing insignificantly or at a maximum number of iterations. The number of unknowns during the joint inversion is the number of prisms for density, plus the number of prisms for magnetization, plus the number of polynomial coefficients, which does not depend on the number of prisms. For example, we would have the number of prisms for both models plus two polynomial coefficients for a linear relationship.

SYNTHETICS EXPERIMENTS

Test 1: compositional and geometrical analysis using a linear relationship

We perform a test on a synthetic model that includes two heterogeneities embedded in a media with linear property variation with depth. In general, we want to gauge the advantages of the CM methodology compared to the conventional separate inversion and to investigate how the gravity and magnetic data, which are comparable in resolution, may influence each other.

Test model and data

We discretize our synthetic model using $22 \times 22 \times 11$ (5324) prisms of 1 km length in horizontal directions and an increasing thickness with depth (Fig. 2). The model is composed of two different homogeneous bodies embedded in a medium with increasing density and magnetization with depth. We assigned a linear relationship between density and magnetization, which results in a straight line in the correspondence map (Fig. 2j). We use 100 km long blocks at the boundaries to account for the contribution of any masses beyond our study area and variable thickness to take advantage of the high sensitivity of shallower prisms and to accommodate topographic features (Fig. 1).

We generated 2500 gravity and 2500 RTP magnetic data in equally spaced $400 \text{ m} \times 400 \text{ m}$ grids, to which we added a Gaussian noise of standard deviation of 2.5 mGal and 7.5 nT, respectively.

Joint inversion process and selection of weighting parameters

In summary, we have seven elements for our joint objective function: misfits for gravity and magnetic data, smoothness for density and magnetization models, correspondence maps function, and density and magnetization *a priori* models (see eq. 1). In the test and application cases, *a priori* models were not used.

The two gravity and magnetic data misfits are weighted by their corresponding standard deviations (2.5 mGal and 7.5 nT). To explore a large initial range of relationships, the joint inversion test started from randomly distributed density and magnetization models instead of conventional homogeneous models, which produce an undesirable one-point cross plot with infinite solutions. Our joint inversion process explores first large smoothness factors, which are gradually reduced until satisfactory misfits and property distributions are found. Simultaneously, the standard deviation of the correspondence map function's effect is gradually reduced (e.g. Lelièvre *et al.* 2012) through iterations to decrease the dispersion around the estimated polynomial function (see Fig. 7).

Separate versus joint inversion

We obtained the separately and jointly inverted models shown in Fig. 2 using the same smoothness factors following the described procedure. The joint inversion algorithm required 12 iterations until the desired RMS misfits were reached. Comparing models recovered from separate and joint inversion, it is noteworthy that the joint procedure shows some improvements. Even though the same smoothness factors were used for separate and joint inversions, in the latter, both properties are distributed on a wider range, and a better bodies' delimitation is achieved.

In the separately inverted models, the horizontal limits of the high density and magnetization volume are revealed unsatisfactorily concerning its top in the density model and its bottom in the magnetization model (Figs 2b and e). In the jointly inverted case, the models appear to be guided by the better sensitivity of gravity data at depth to delimit slightly improved the bottom of the body (Figs 2c and f), and the suitable shallower sensitivity of magnetic data to delineate the top of the body. Enhanced recovery is also reached at lower density and magnetization contrasts (see the right body in Fig. 2). In general, recovered bodies fit the true anomalous bodies better with respect to values and geometry in joint inversion. This is best reflected in the reduction in the scattering of the correspondence maps to a minimum and the expansion of the range of values (Fig. 2l). Note that the linear trend of original models (Fig. 2j) was not recovered in the separate inversions (Fig. 2k). In the joint inversion, the slope is similar to the true slope, but the intercept is slightly shifted. We believe this is caused mainly by the smoothness conditions, which do not allow the recovery of the extreme values of the properties and so defining the intercept. Because of the poor vertical resolution of potential data, neither inversion was able to recover the depth trend.

To assess the compositional improvements, we show the absolute values of residuals comparing the jointly and separately inverted models with the true models (Fig. 3). The residual maps highlight the differences which are: (i) the density values for the anomalous bodies are closer to the true values using joint inversion (compare Figs 3a, e, i versus c, g, k), (ii) the magnetization values also are closer in the jointly inverted models, although the differences with the separately inverted magnetization models are not as notorious as were in the density case (compare Figs 3b, f, j versus d, h, l), (iii) the space above anomalous bodies (between 0 and 1.5 km) shows smaller residuals in the jointly inverted density models in comparison with separately inverted models (see Figs 3a, e versus c, g), resulting in a better delimitation of the top of the body, (iv) the space below the anomalous bodies (4.2–8.25 km) has smaller differences in the jointly inverted magnetization models in comparison with the separately inverted ones (compare Figs 3b, f with d, h), leading to a better delimitation in the bottom of the bodies and (v) finally, jointly inverted models show smaller differences with the original background medium in comparison with separately inverted models, especially in the magnetization values (compare Figs 3j with l).

To evaluate the geometrical improvements we calculated the horizontal spatial gradients $\frac{\partial m_1}{\partial x}$ (Figs 4a, as, aj), $\frac{\partial m_2}{\partial x}$ (Figs 4b, bs, bj), $\frac{\partial m_1}{\partial y}$ (Figs 4c, cs, cj) and $\frac{\partial m_2}{\partial y}$ (Figs 4d, ds, dj) for true, separately and jointly inverted models, respectively. We use the square root of absolute values to reduce its range and highlight the differences. In general, the horizontal gradients of jointly inverted models are closer to the horizontal gradients of synthetic models, especially true in the density case (Figs 4aj and cj) compared to separately density inverted models (Figs 4as and cs).

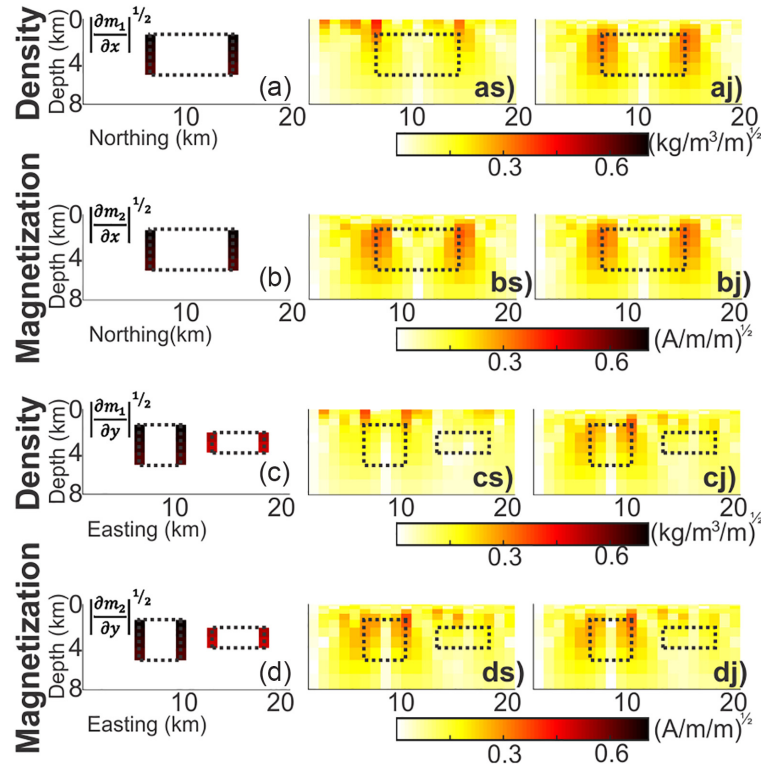


Figure 2. Comparison of separately and jointly inverted models. The true density and magnetization distributions followed by the distributions from separate and joint inversion are displayed in the three columns from left- to right-hand side. They are illustrated along centred planes in (a–c) XZ , (d–f) YZ and (g–i) XY . (j–l) Correspondence maps for density versus magnetization obtained from the true model, as well as from separate and joint inversion after 12 iterations with smoothness factors $\alpha_1 = 5$, $\alpha_2 = 5$, $\sigma_{d1} = 2.5$ mGal, $\sigma_{d2} = 7.5$ nT and $\sigma_{gg} = 0.05$.

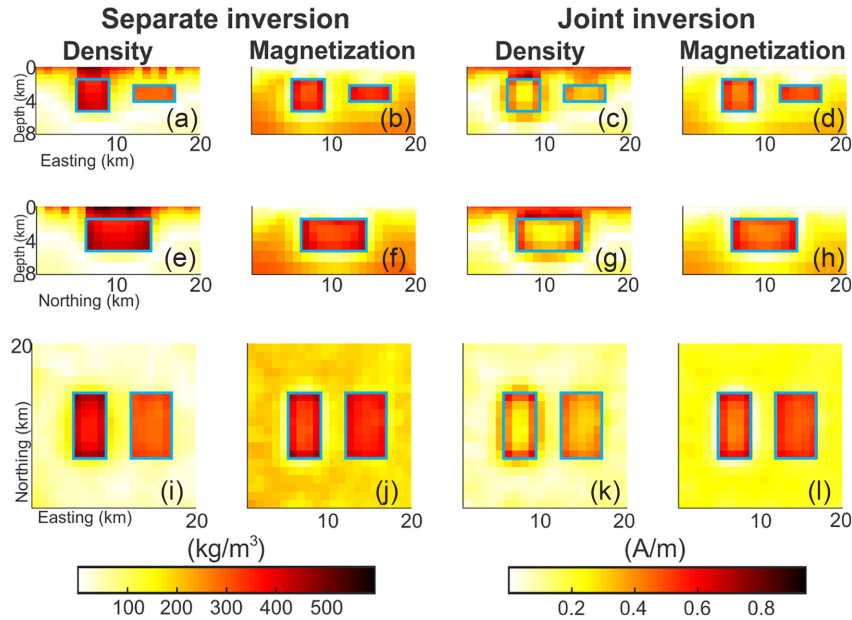


Figure 3. Residual maps of separately and jointly inverted models compared with the true synthetic models along the centred planes XZ (a–d), YZ (e–h) and XY (i–l). Blue lines show the original position of anomalous bodies.

Fig. 5 shows the steady development of the data misfit through the iterations. As described above, the algorithm starts from a random distribution of density and magnetization at the initial iteration (Figs 5b and e), a weak direct coupling relationship, and smooth

models (Fig. 5h). The RMS misfit of gravity and magnetic data decreases significantly in the first four iterations, after which it reaches the expected weighted data misfit RMS ≈ 1 (Fig. 5a). At this fourth iteration, the correspondence map function's coupling is still weak,

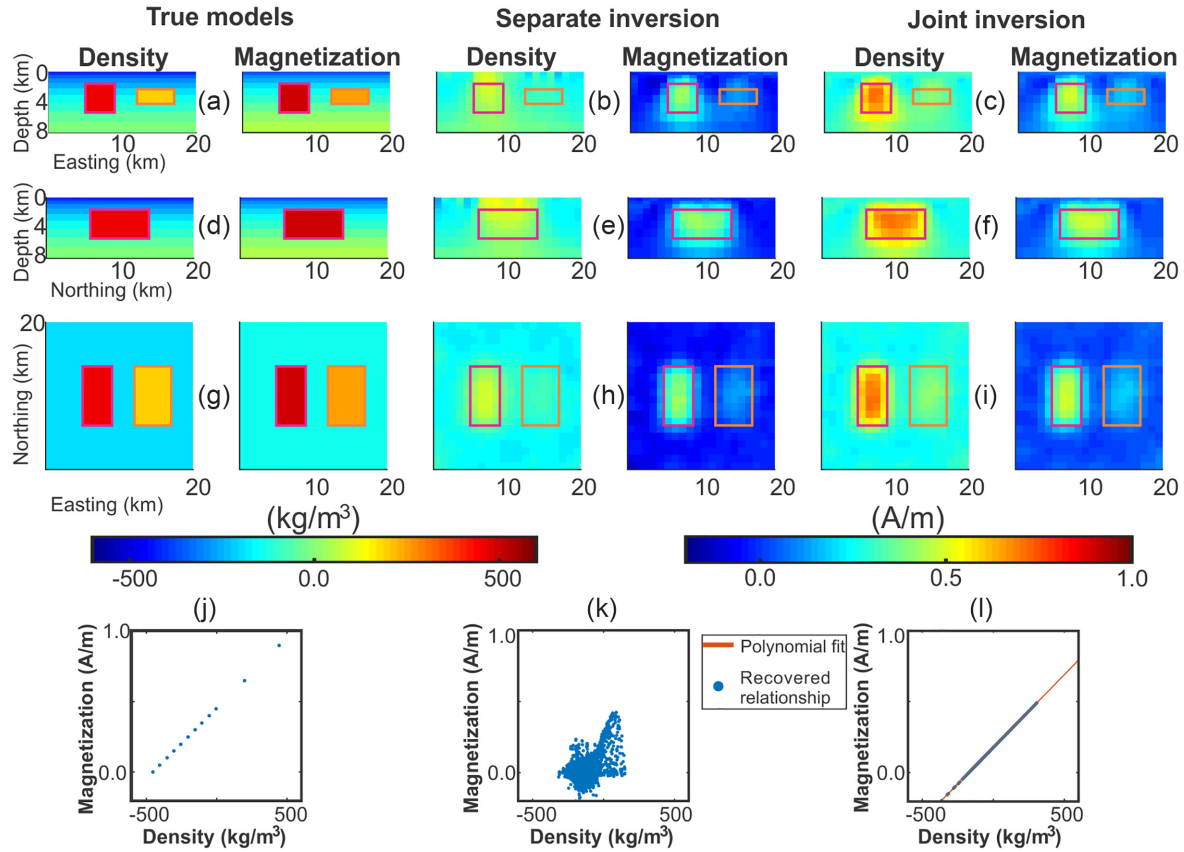


Figure 4. Spatial gradients of separately and jointly inverted models through horizontal directions shown at different slices. The values $|\frac{\partial m_1}{\partial x}|^{\frac{1}{2}}$ are shown in (a) as (aj) for true, separately and jointly inverted models. The values $|\frac{\partial m_2}{\partial x}|^{\frac{1}{2}}$ are shown in (b), (bs), (bj) for true, separately and jointly inverted models. Similarly, for y direction, the values $|\frac{\partial m_1}{\partial y}|^{\frac{1}{2}}$ and $|\frac{\partial m_2}{\partial y}|^{\frac{1}{2}}$ are shown in (c), (cs), (cj) and (d), (ds), (dj) for density and magnetization, respectively. Black dotted lines show the original position of anomalous bodies.

which is evident in the bad fitting to the linear trend (Fig. 5i). It seems that at this step, the recovered models still preserve the features of separate inversion. Starting from iteration #4, the algorithm focus on increasing the correspondence map function contribution, reaching a linear fit at iteration #12 (Fig. 5j). RMS data misfits increases slightly from separate to joint inversion (see Fig. 6) from 0.88 to 0.95 (gravity) and 0.80–0.82 (magnetic). Nonetheless, both are in the expected range. The RMS of the correspondence map fit decreases from 36 in iteration #1 to 6 in iteration #12.

Fig. 7 shows the relative change in models compared with the solution in the previous iteration. Polynomial coefficients are the most unsteady parameters in contrast with density and magnetization. The influence of smoothness is gradually reduced while the influence of CM's function is gradually increased until $\alpha_1 = \alpha_2 = 5$ and $\sigma_{gg} = 0.05$, respectively.

Test 2: parabolic relationship in a potential geothermal scenario

The second test aims at recovering a study model that contains some of the geophysical features observed in the geothermal scenario in Los Humeros, México and others. The centre of the synthetic model is composed of low-density resembling volcano-sedimentary filling material hosted by high magnetic andesites. Neighbouring, there are four structural controls resembling faults regions characterized

by low density—low magnetization values (e.g. Miller & Williams-Jones 2016). The hydrothermal alteration, commonly controlled by the structural features, produces demagnetization processes creating a transition area (e.g. Airo 2002; Clark 2014). Finally, two deeper intrusive bodies with high density and moderate magnetization are located between the centre of the caldera and the faults zones (e.g. Zhdanov *et al.* 2012b). The density and magnetization models drawn on a cross-property plot shapes a parabolic relationship. This relationship will allow us to explore a more complex scenario than the previous one and test a model forward to the field case. Figs 8(a), (d) and (g) show the synthetic model in vertical and horizontal sections and its cross-plot.

We discretize our synthetic model using $23 \times 23 \times 11$ (5819) prisms of 1 km length in horizontal directions and an increasing thickness with depth. We created 7056 gravity and 7056 RTP magnetic data in equally spaced $250 \text{ m} \times 250 \text{ m}$ grids, which were contaminated with Gaussian noise of standard deviation of 2.5 mGal and 7.5 nT, respectively. The joint inversion setting was the same as the used in the previous test. The only difference is that CM's standard deviation and smoothness factors were fixed instead of gradually reduced through iterations as done in the field case.

Fig. 8 shows horizontal slices at a depth of 1.5 km and vertical sections to compare the separately and jointly inverted model. The separately obtained magnetization model recuperates the main features: the low magnetization zones representing fault regions, the central high magnetization body from separate inversion, and

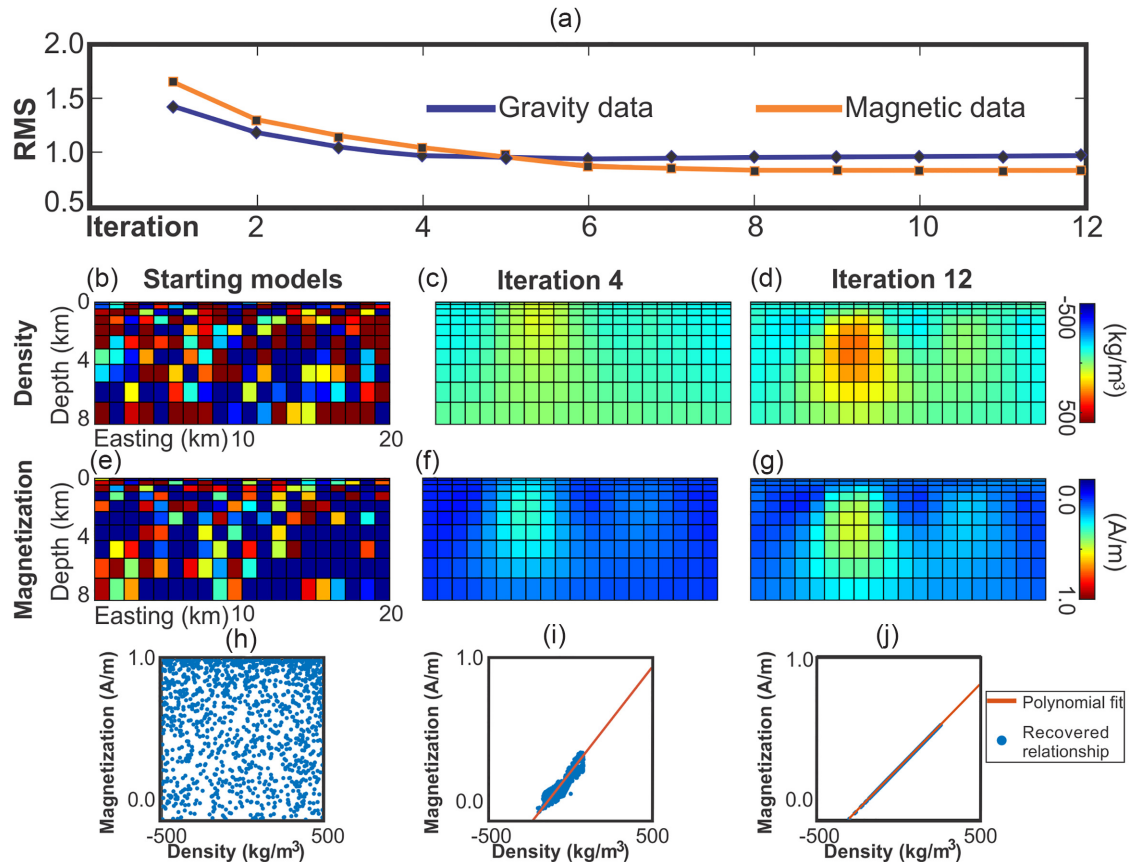


Figure 5. Evolution of the inversion process illustrated by (a) the RMS data fit for the gravity and magnetization model, (b–g) three different iteration steps (1st, 4th and 12th) of the density and magnetization distribution along the central plane in XY direction with a random distribution at in the initial (left-hand column) and the final distribution at iteration 12 (right-hand column). (h–j) Correspondence maps of the three steps with increasing fitting to a line from the starting model (left-hand column) to the final iteration 12 (right-hand column).

the hydrothermal alteration low magnetization zone. Although the separately obtained density model roughly recovers the different bodies, they are blurred, and their limits are not clear. For example, the low-density zones representing fault zones are almost imperceptible (see Fig. 8h).

The jointly inverted magnetization model shows a slight improvement in the recovered values. However, the main difference is found in the density model. The four low-density zones representing the fault zones are now satisfactorily delimited as well as the low-density caldera centre and the high-density intrusive bodies. Despite this focusing effect on the different bodies, the density values were not recovered as good as the magnetization values.

Linear and cubic relationships

In the previous case, we explored the possibility of recovering a relatively complex scenario using a parabolic polynomial. However, when faced with field data, we do not know in advance which is the potential polynomial that best fits the density–magnetization relationship. Therefore, an inevitable research question is what occurs if we use different relationships. We performed the same joint inversion process using a linear relationship and using a cubic relationship. Fig. 9 shows the results in horizontal slices from models at a depth of 1.5 km, and Table A1 (Appendix A) collects the RMS misfits and iterations required for each experiment.

In the first case, the linear relationship (Figs 9a and b) took four iterations until models were insignificantly changing. As seen in Fig. 9, the density model recovered just the lowest values. High-density intrusive bodies are almost imperceptible.

The cubic relationship (Figs 9c and d) was unable to achieve the CM's misfit criteria and stopped when it reached the user's criteria (maximum 30 iterations, at CM's RMS = 2.4082). We took the last iteration to analyse the results. The obtained models recuperated better the structures than the linear case, and the final relationship is similar to the parabolic case. Fig. 9 shows the cubic relationship fitting through iterations. At the first iterations, we observe the curve containing two concaves (Fig. 9e) in the range of models. Slowly it becomes a curve similar to the parabolic relationship shifting one of the concaves out of the range of values (Figs 9e and f). This could indicate to us that we can use a higher polynomial than the required obtaining acceptable outcomes. However, the real cost of using a higher-grade polynomial is the spent time (30 iterations) and the CM's misfit produced by the third-grade coefficient. Moreover, the stability of the coefficients' solution could be compromised. Fig. 9(h) shows the relative change in coefficients at each iteration (comparing the updated solution with the previous one). Linear (in red) and parabolic (in green) polynomials reveal a general downward trend, while the cubic (in blue) polynomial's coefficients show an unstable behaviour with relative changes going up and down.

These two examples of enforcing the relationships to be fitted using inappropriate polynomials show that, although it is not possible

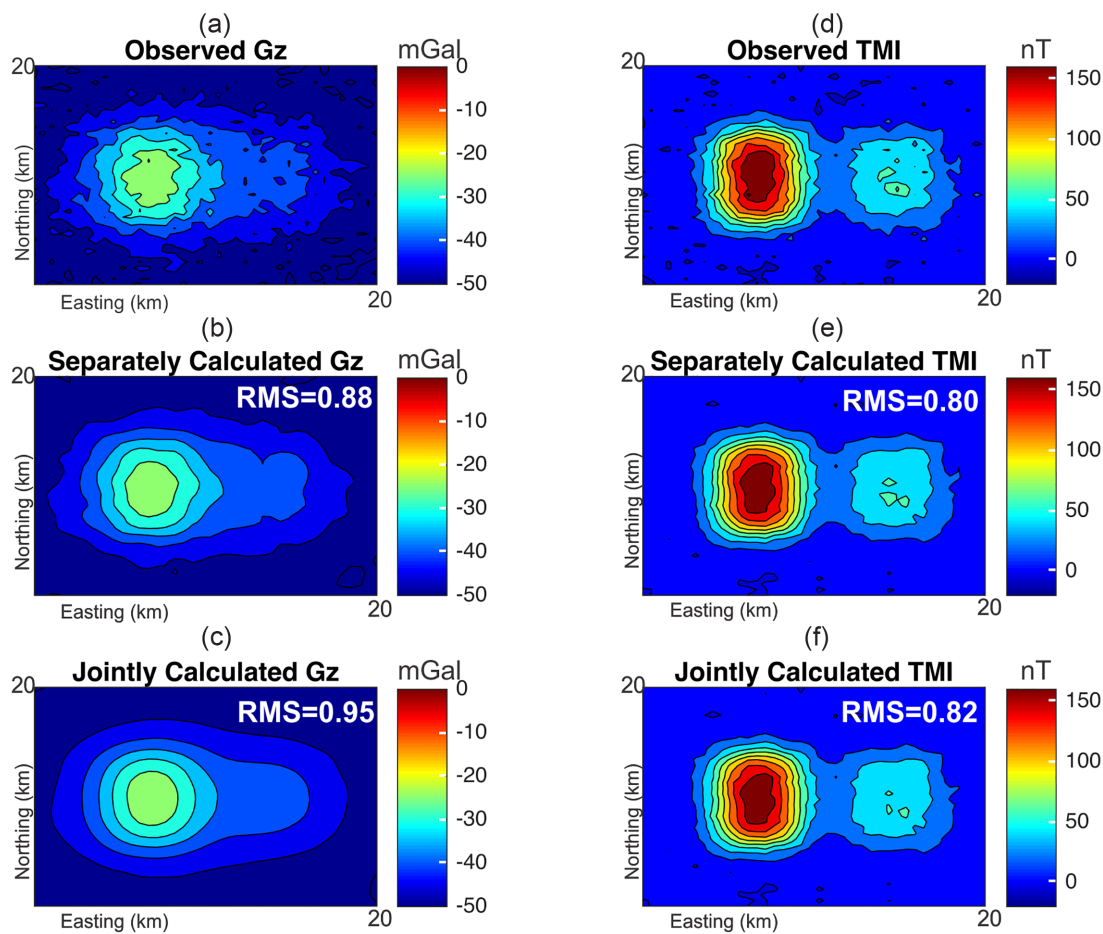


Figure 6. Panels (a) and (d) are the observed vertical gravity (Gz) and the total magnetic intensity (TMI) contaminated with 2.5 mGal and 7.5 nT, respectively, panels (b) and (e) are Gz and TMI computed from separate inversion and panels (c) and (f) are Gz and TMI computed from joint inversion.

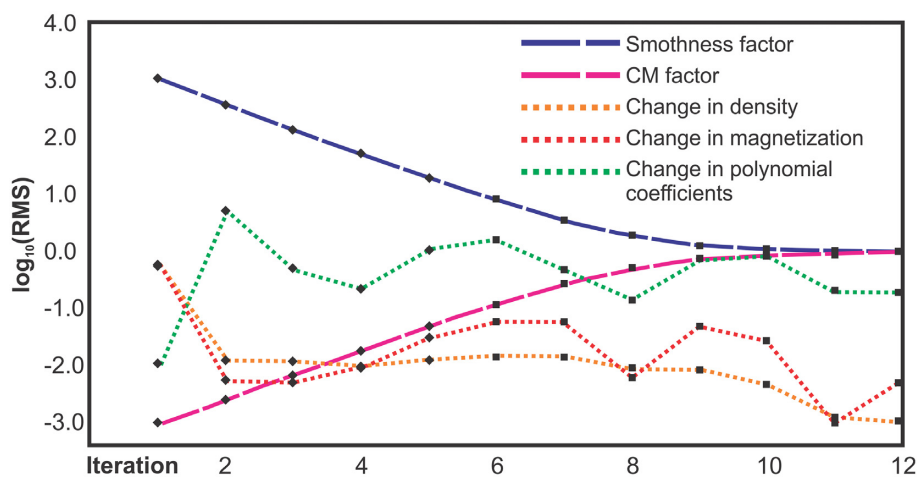


Figure 7. The evolution of the inversion process is illustrated by (i) the change in density, magnetization and polynomial coefficients and (ii) the smoothness and CM' multipliers of the factors used in the separate inversion. These multipliers are dynamically introduced with iterations from large values in the case of the smoothness and from small values in the case of the CM' function. The final CM' standard deviation reached is $\sigma_{gg} = 0.05$ and the final smoothness factors reached are $\alpha_1 = \alpha_2 = 5$.

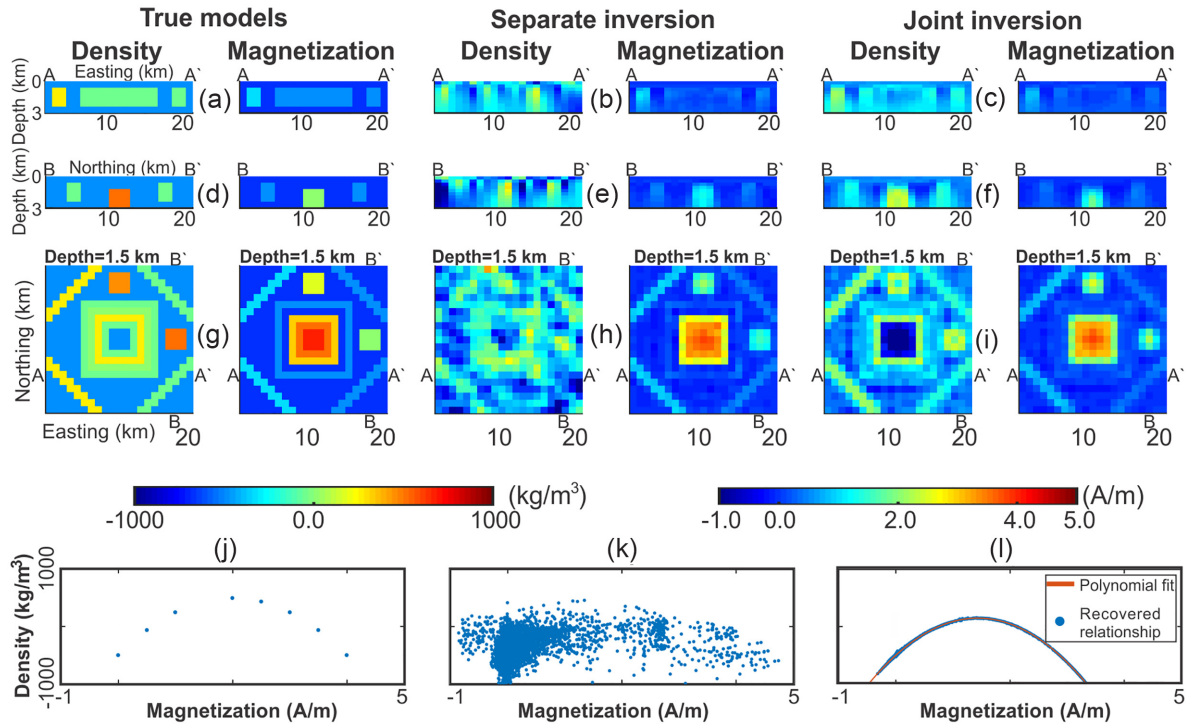


Figure 8. (a, d) True density and magnetization vertical sections; (g) true density and magnetization horizontal section; (b, e) separately inverted density and magnetization vertical sections; (h) separately inverted density and magnetization horizontal section; (c, f) jointly inverted density and magnetization vertical sections; (i) jointly inverted density and magnetization horizontal section and (j, k, l) cross-property maps.

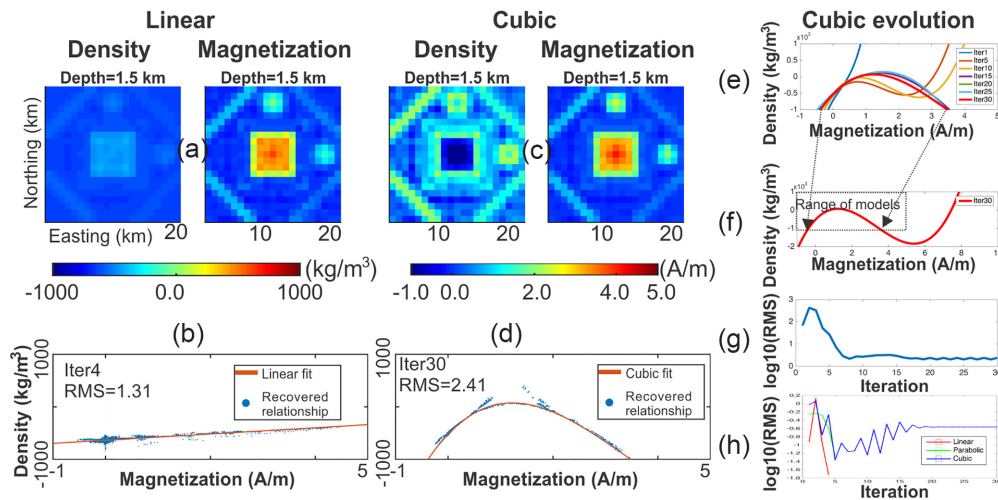


Figure 9. Panels (a) and (b): jointly inverted models using a linear relationship. Panels (c) and (d): jointly inverted models using a cubic relationship. Panels (e)–(g): evolution of the cubic relationship through iterations and its CM's RMS misfit. Panel (h): the relative change in coefficients through iterations comparing the linear (4 iterations), parabolic (6 iterations) and cubic (30 iterations) polynomials.

to select the polynomial grade automatically, the RMS data misfit is a reliable indicator of the relationship's consistency.

The achieved linear relationship overfitted magnetic data (0.93) and underfitted gravity data (1.25). This can be explained considering that we have a non-unique relationship. While the magnetization range can be covered using one positive linear trend (0–5 A m⁻¹), the density needs one positive trend (–500 to 500 kg m⁻³) and one negative trend (500 to –500 kg m⁻³). The unrecovered gravity data's values in one of the trends produced the underfitted gravity data,

while the well-covered magnetization range produced the overfitted magnetic data.

In the case of the cubic relationship, the magnetic and gravity misfits are comparable to those obtained through the parabolic relationship ($\text{RMS}_{\text{gravity,cubic}} = 1.0274$, $\text{RMS}_{\text{gravity,parabolic}} = 1.0306$ and $\text{RMS}_{\text{magnetic,cubic}} = 0.9681$, $\text{RMS}_{\text{magnetic,parabolic}} = 0.9699$) and this is translated in a better recovering of property's values than the linear case. However, it took more iterations, and the final polynomial containing just one concave in the range of recovered values reveals

that it is not necessary that extra polynomial grade. Moreover, the parabolic relationship case reached a better CM's fitting and took only six iterations.

APPLICATION TO LOS HUMEROS GEOTHERMAL FIELD, MEXICO

Gravity and magnetic data modelling has been widely applied to geothermal targets worldwide for structural and physical characterization (e.g. Nunziata & Rapolla 1981; Altwegg *et al.* 2015).

Many geothermal fields accommodate their laterally confined reservoirs into layered sedimentary or metamorphic sequences (e.g. Şimşek 1985; Batini *et al.* 2003; Geiermann & Schill 2010). Here, our CM joint inversion is applied to gravity and magnetic data from the Los Humeros Volcano Complex (LHVC) that includes the Los Humeros conventional and super-hot geothermal fields.

The conventional target of the Los Humeros geothermal field is mostly located in the pre-caldera volcanic layers (mainly andesite). Super-hot fluids are expected to occur in the underlying carbonates of the meta-sedimentary basement. The ignimbrites forming the caldera seem to act as a seal-rock (Carrasco-Núñez *et al.* 2017). In the basement and pre-caldera rocks, the multistage Pleistocene silicic volcanism led to two major episodes of caldera collapse (Norini *et al.* 2015): Los Humeros Caldera (21 km × 15 km) and Los Potreros Caldera (9 km diameter).

To characterized the structures of the LHVC from aeromagnetic data, Flores-Luna *et al.* (1978) identified a regional magnetic dipolar anomaly associated with a highly magnetized body over Los Humeros Caldera from aeromagnetic data. Campos-Enriquez and Garduno-Monroy (1987) applied 2-D forward modelling to set the contact between volcano-sediments and limestones along with an E–W profile. All this previous information reveals a complex scenario that may result in non-linear density–magnetization relationships. We apply our CM joint inversion to determine the overall structure and the density–magnetization relationship of the LHVC. We then discuss their implications for geothermal characterization.

Data

A Bouguer anomaly was calculated from gravity data of 1001 gravity stations provided by the Comisión Nacional de Hidrocarburos (<http://www.gob.mx/cnh>) using a Bouguer density of 2670 kg m⁻³. We subtracted a regional trend using a second-order polynomial fit, the regional trend was subtracted from this Bouguer anomaly to obtain a suitable residual anomaly for the studied area (Fig. 10a). In total, 1001 stations were used to obtain the residual Bouguer anomaly map.

5165 data from a regional aeromagnetic survey were provided by the Servicio Geológico Mexicano (<http://www.sgm.gob.mx/>). This survey was flown at 300 m above ground level along 1-km-spaced N–S flight lines with E–W tie lines every 5 km. The data underwent through aircraft, tidal, IGRF and microleveling corrections; we then subtracted the overall mean as a regional magnetic field to furnish the total magnetic intensity (TMI) data used for our inversion experiment (Fig. 10d). Based on the IGRF 2000, the assumed direction of the inducing geomagnetic field is $D = 5.37^\circ$ and $I = 48.16^\circ$.

The largest gravity minimum extends over the central and NE part of the Los Humeros caldera and barely reaches the production zone at the SW. This gravity anomaly appears to belong to a NE–SW trending chain of negative anomalies that crosses the caldera and its rim. Similarly, a large dipolar magnetic anomaly extends across the

entire caldera, including the production zone. Magnetic remanence was not considered.

Model parametrization

In the following, we describe the discretization and regularization parameters (smoothness and correspondence maps factors) that were used for our joint and separate inversions of the LHVZ data.

A 30 km × 30 km area was divided into 1 km (N–S) × 1.5 km (E–W) rectangular compartments. Each of these compartments extends from the topographic relief to a depth of 4.8 km below sea level and accommodates 20 rectangular prisms with thicknesses increasing from 0.1 m down to a maximum of 1100 m at the maximum depth. Laterally, the padding cells are extended to 100 km to account for lateral effects. Comparable to the test experiment, *a priori* density, and magnetization models m_{01} and m_{02} were not used. The search process starts from random density and magnetization (with uniform density ranging from -1 to 1).

We performed individual inversion of each data set to establish a reference misfit and tune appropriate regularization parameters useful for the joint inversion (Moorkamp 2017). The current joint inversion approach requires a carefully balanced weighting for the elements of the objective function (eq. 1), which we determine by trial and error from the separate inversions. The data standard deviations are normally associated with the specific surveying strategy; here, we assume standard deviations of 0.1 mGal and 20 nT for gravity and magnetic data, respectively. Smoothness factors of $\alpha_1 = 50$ and $\alpha_2 = 5$ were used for the density contrast and magnetization models, respectively.

In the case of joint inversion, a standard deviation for the correspondence map function of $\sigma_{gg} = 5$ per cent was applied. We started the search for the coefficients describing the polynomial in the CM's function from the simplest case: a linear relationship between density and magnetization. Then we added terms in the polynomial until we reached a satisfactory fitting in the correspondence maps. Table B1 (Appendix B) shows the RMS misfit for each experiment. The cubic experiment diverged at iteration 16. The best fitting in gravity and CM's function was achieved by the parabolic relationship, while the linear relationship achieved the best fitting in magnetic data. However, we believe this effect over magnetic data is similar to the test case, where only the magnetization range was correctly covered, favouring the magnetic data fitting over the gravity data fitting. Considering the best global fitting, we selected the quadratic relationship as the most suitable, which was also consistent with the previous geological information.

Convergence of separate and joint inversion

Our joint inversion algorithm reaches a suitable RMS after 12 iterations (Fig. 11). Comparably to the test case, the final RMS data misfit values slightly increased from 6.21 to 6.59 and 1.81 to 2.22 for the density contrast and magnetization models when applying separate and joint inversion, respectively. Similar to the test case, the RMS data fitting level reached in separate inversion was reached by joint inversion after the first four iterations even though we adopted fixed values for weighting smoothness and correspondence map function instead of dynamically relax the smoothness values and increase the correspondence map function contribution. After the fifth iteration, the algorithm improves the coupling of the correspondence map function. The RMS correspondence map fit decreases from 56 to 9.9 and 5.0 at iterations #1, #5 and #12, respectively. Finally, a

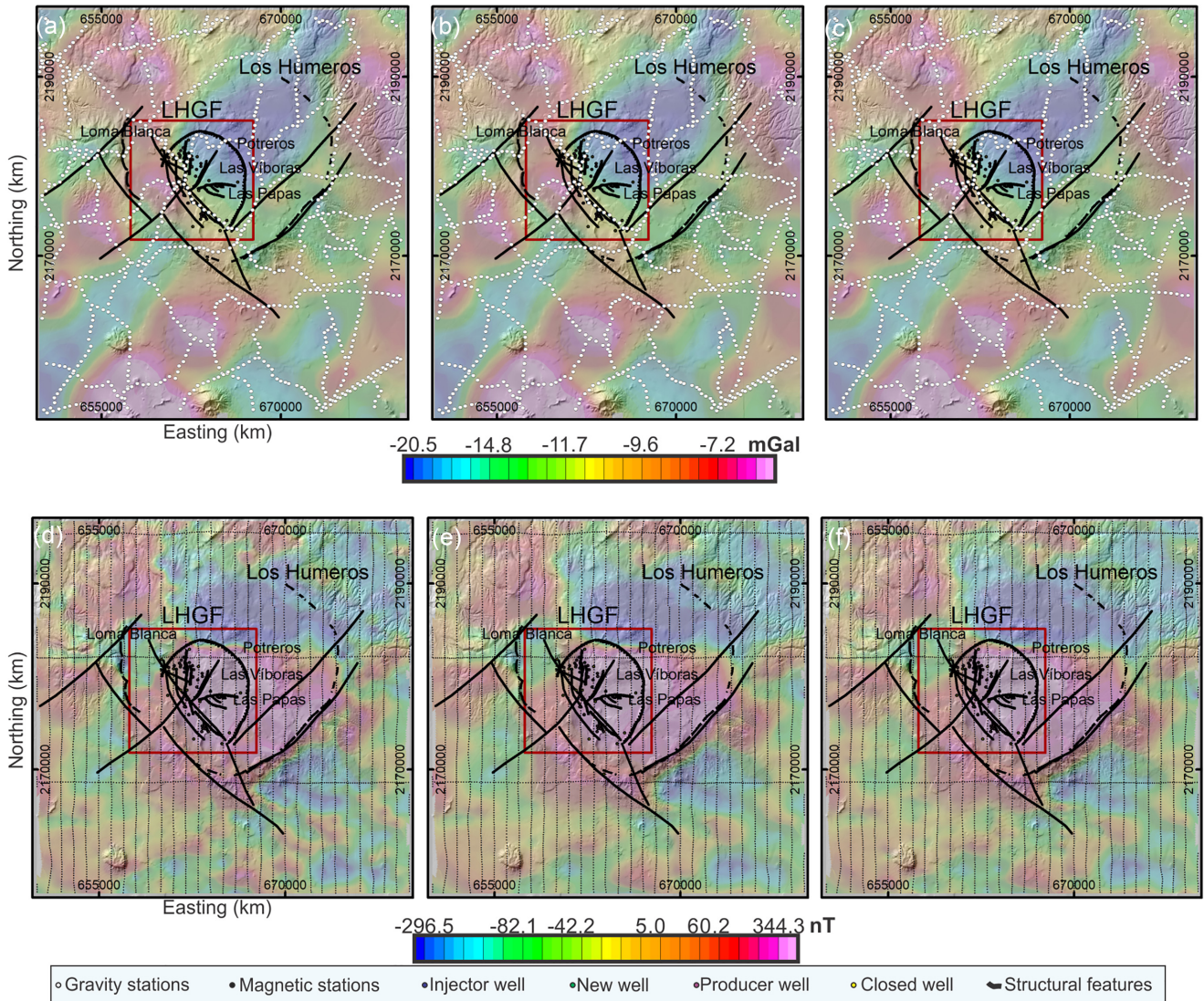


Figure 10. (a) Residual map of the Bouguer anomaly of the Los Humeros area (data source: Comisión Nacional de Hidrocarburos, <http://www.gob.mx/cnh>). Measurements were acquired along the main roads. Predicted Bouguer anomaly data from (b) the separate inversion and (c) joint inversion. (d) Total magnetic field anomaly of the Los Humeros area obtained from aeromagnetic data (data source: Servicio Geológico Mexicano, <http://www.sgm.gob.mx/>). Measurements are acquired along N–S lines with an offset of 1 km and E–W lines with an offset of 5 km at 300 m flight altitude. Predicted magnetic field from (e) separate inversion and (f) joint inversion. The red rectangle shows Los Humeros Geothermal Field (LHGF) area.

parabolic relationship between density and magnetization is fitted in the correspondence map (Fig. 11j). The largest scattering in the cross plot occurs at the high magnetization values, which seems to invert the positive trend shown at the lower values, producing a concave function between density and magnetization relationship. We may distinguish two main zones in the correspondence map divided by the vertex point in the parabola, one containing low density and magnetization values and the second one containing low density and high magnetization values.

Analysis of jointly inverted models

The separately estimated regional density and magnetization bodies show good lateral resolution but broad tails at depth, which are reduced in extent in the joint inversion (see Figs 12a and b versus c and d). While the lateral limits of regional structures seem to

be recovered in both cases, jointly inverted models show focused bodies at depth.

The results of the joint inversion are presented along centred E–W and N–S profiles and horizontal slices (Figs 12d, e and f, respectively) and compared to major faults inside or delimiting the caldera. The fault zones indicated in the E–W profile (Fig. 12d) comprise a region where comparably low magnetization (between 1 and 2 A m⁻¹) is dominant. This seems to be characteristic of N–S oriented major faults and, given the volcanic environment, may indicate hydrothermal alteration along the faults (e.g. Airo 2002; Miller & Williams-Jones 2016). In particular, Los Humeros caldera is closely related to low magnetization values (Fig. 12f). Simultaneously, in the density contrast model, the same fault zones are characterized by comparably low density. To the east of the geothermal field, density contrast is characterized by positive values. Los Potreritos faults reveal a density contrast of -400 kg m^{-3} . Such density contrasts along fault zones have been attributed to fracture porosity in other geothermal fields (Altwegg *et al.* 2015).

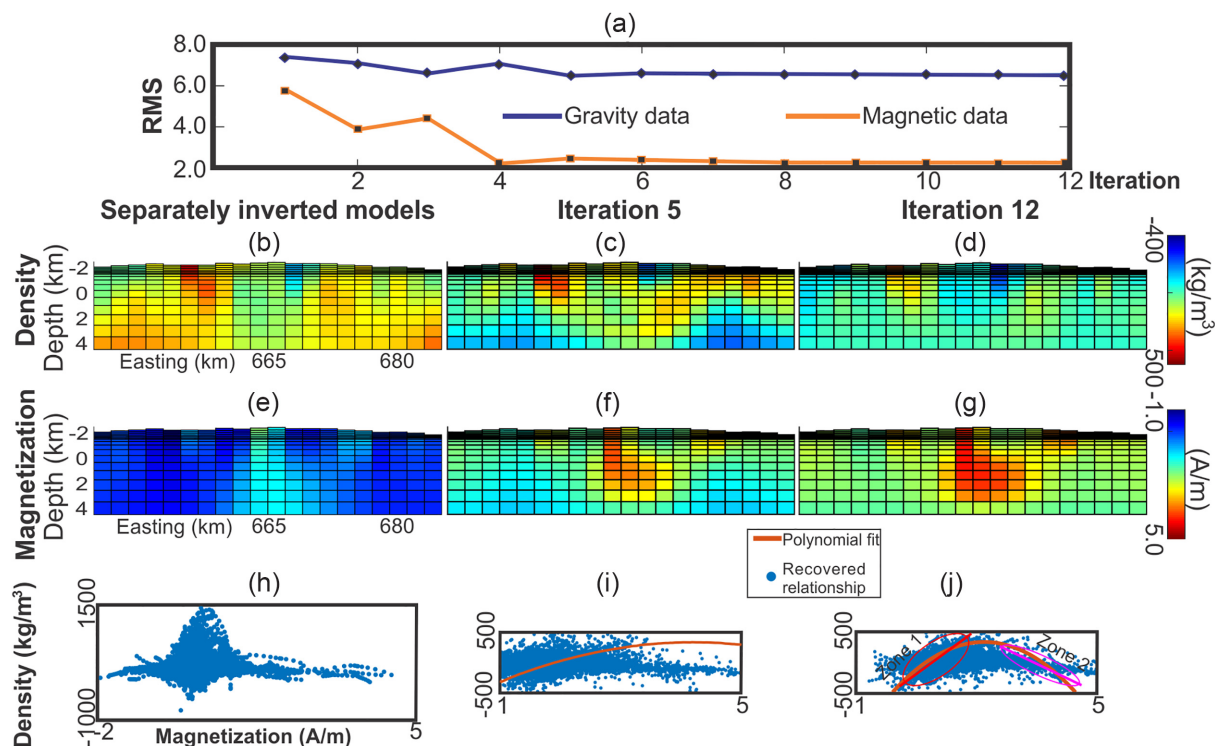


Figure 11. Evolution of the joint inversion process illustrated by (a) the RMS data fit for the gravity and magnetization model, (b, e) the density and magnetization models from separate inversion and (h) its cross-plot. Joint inversion of the three representative steps with increasing fitting to a polynomial function from the iteration #5 (c, f), and to the final iteration #12 (d, g), and the corresponding density–magnetization cross-plots (i, j).

The second zone encompasses a major low-density anomaly, and an area of high magnetization appears in the central to the northern part of the caldera and the east of the Loma Blanca Fault. This fault and the subparallel minor faults limit the production zone at the west of the caldera (Figs 12d–f). Compared to the results from separate inversions, where mainly vertical to subvertical boundaries appear, in particular, the density contrast distribution resulting from the here presented property-driven inversion reveals the influence of the subhorizontal basement limestone unit reported earlier by Campos-Garduño & Arredondo-Fragoso (1992) and Árzate *et al.* (2018). While the density anomaly appears more diffuse, the high magnetic anomalous body is clearly delimited, particularly at depth. It reproduces the large dipolar anomaly observed in the magnetic map (Fig. 10d). The magnetic anomaly pattern in LHGF: the central high magnetic and the annular low magnetic following Los Humeros caldera (Fig. 12f) supports the idea of the existence of hydrothermal alteration (see Clark 2014).

CONCLUSIONS

We have extended the CM methodology to jointly inverting gravity and magnetic data for 3-D targets. The approach has confirmed to improve the inversion results compared to separate inversion. The numerical assessment comparing separately and jointly inverted models demonstrated that some compositional and geometrical improvements are achieved. The gradual property variation favours the detection of multiple subzones (heterogeneities), and thus, a more accurate integrated subsurface image is completed at the expense of a slightly increased RMS misfit value. The parabolic relationship synthetic experiment allowed us to explore the behaviour of CM when using different polynomials and confirmed its ability to

properly recover potential geothermal structures using gravity and magnetic data. Weighting the elements in the objective function is accomplished by defining the correspondence maps' influence iteratively in the field case and the parabolic relationship experiment or dynamically relaxing the smoothness and increasing the contribution of correspondence maps' function in the linear relationship experiment. In practice, the influence of the five elements in the objective function is reduced to the determination of the correspondence map's weighting because the two smoothness factors are taken from the separate inversions and the two sets of standard deviations directly from the data. The stability analysis results showed that the polynomial coefficients are the most unsteady parameters during inversion. The effects of the instability are particularly critical when we use higher grade polynomials than required. In the second experiment, we explored that case by using higher and lower-grade polynomials. The search was managed adequately by gradually increasing the number of coefficients until the proper RMS misfits were achieved. The strategy used in the second experiment was successfully implemented to the field data, which confirms its applicability.

The application of our joint inversion approach to the Los Humeros case conforms with a quadratic density–magnetization distribution fitting the geological setting of the region, including, in the case of density, the subhorizontal basement structures. Of utmost importance for geothermal exploration, however, is the striking link between low density, low magnetization, and fault zones. These zones could be related to hydrothermal alteration such as previous authors have reported in other geothermal systems. Note that this link is not evident from separate inversion. Although gravity and magnetic data indicate the co-existence of major magnetization and density anomalies, significant differences in the spatial extents are brought out in the distributions resulting from this joint inversion.

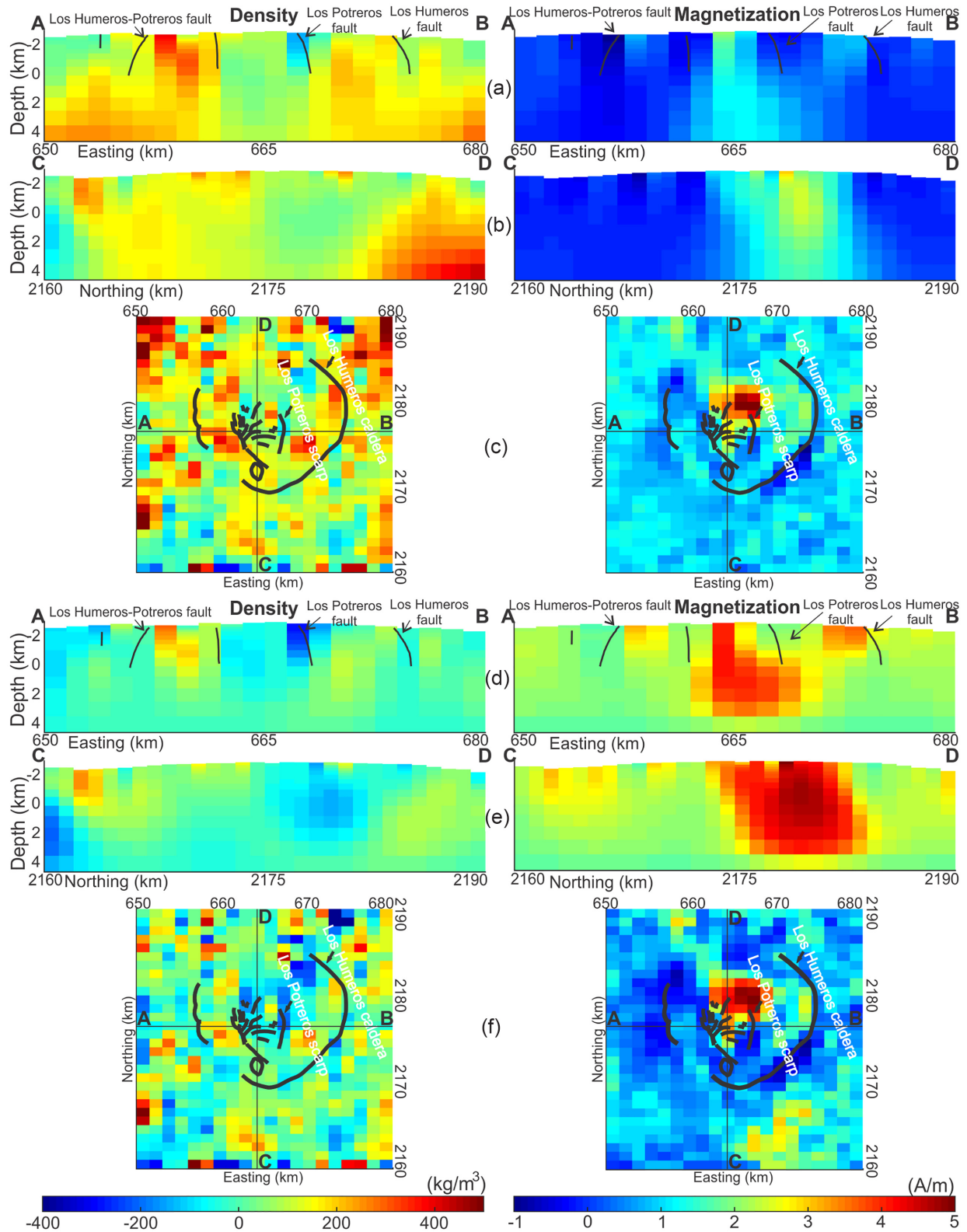


Figure 12. Comparison between separately and jointly inverted models. Density contrast (left-hand panels) and magnetization (right-hand panels) of the separate (a–c) and joint (d–f) inversions are displayed at different perpendicular profiles at $X = 665$ km, $Y = 2178$ km UTM and $Z = 1$ km a.s.l.

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DATA AVAILABILITY STATEMENT

The data underlying this article were provided by Comisión Nacional de Hidrocarburos (CNH) and Servicio Geológico Mexicano (SGM). Data will be shared on request to the authors with permission of CNH, SGM and the GEMex project. The source code will be shared on reasonable request to the corresponding author.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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APPENDIX A

Table A1.

Table A1. Linear, parabolic and cubic polynomials RMS misfit comparison in test 2.

Polynomial	Iterations	Gravity RMS misfit	Magnetic RMS misfit	CM's RMS misfit
Linear	4	1.248	0.927	1.312
Parabolic	6	1.031	0.970	1.053
Cubic	30	1.027	0.968	2.408
Separate		0.957	0.874	

APPENDIX B

Table B1.

Table B1. Linear, parabolic and cubic polynomials RMS misfit comparison in the field case.

Polynomial	Iterations	Gravity RMS misfit	Magnetic RMS misfit	CM's RMS misfit
Linear	15	8.129	1.818	7.195
Parabolic	12	6.599	2.225	4.987
Cubic	15	7.165	2.695	47.138
Separate		6.213	1.814	